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Advanced Math

15

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ANSWER KEY · SAT QUESTION BANK

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Q1

51

The correct answer is 51. A quadratic equation of the form $ax^2 + bx + c = 0$, where a, b, and c are constants, has no real solution if and only if its discriminant, $-4ac + b^2$, is negative. In the given equation, $a = -1$ and $c = -676$. Substituting -1 for a and -676 for c in this expression yields a discriminant of $b^2 - 4 \cdot 1 \cdot 676$, or $b^2 - 2{,}704$. Since this value must be negative, $b^2 - 2{,}704 < 0$, or $b^2 < 2{,}704$. Taking the positive square root of each side of this inequality yields $b < 52$. Since b is a positive integer, and the greatest integer less than 52 is 51, the greatest possible value of b is 51.

Q2

C

C. 970

Choice C is correct. It's given that a quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. This quadratic function can be defined by an equation of the form $f x = a x - h^2 + k$, where $f x$ is the height of the object x seconds after it was launched, and a, h, and k are constants such that the function reaches its maximum value, k, when $x = h$. Since the model indicates the object reaches its maximum height of $1{,}034$ feet above the ground 8 seconds after being launched, $f x$ reaches its maximum value, $1{,}034$, when $x = 8$. Therefore, $k = 1{,}034$ and $h = 8$. Substituting 8 for h and $1{,}034$ for k in the function $f x = a x - h^2 + k$ yields $f x = a x - 8^2 + 1{,}034$. Since the model indicates the object has an initial height of 10 feet above the ground, the value of $f x$ is 10 when $x = 0$. Substituting 0 for x and 10 for $f x$ in the equation $f x = a x - 8^2 + 1{,}034$ yields $10 = a 0 - 8^2 + 1{,}034$, or $10 = 64a + 1{,}034$. Subtracting $1{,}034$ from both sides of this equation yields $64a = -1{,}024$. Dividing both sides of this equation by 64 yields $a = -16$. Therefore, the model can be represented by the equation $f x = -16 x - 8^2 + 1{,}034$. Substituting 10 for x in this equation yields $f 10 = -16 10 - 8^2 + 1{,}034$, or $f 10 = 970$. Therefore, based on the model, 10 seconds after being launched, the height of the object above the ground is 970 feet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

B

B. $-3 < x < 1$

Choice B is correct. The graph of a quadratic function in the xy-plane is a parabola. The axis of symmetry of the parabola passes through the vertex of the parabola. Therefore, the vertex of the parabola and the midpoint of the segment between the two x-intercepts of the graph have the same x-coordinate. Since $f(-3) = f(-1) = 0$, the x-coordinate of the vertex is $\frac{(-3)+(-1)}{2} = -2$. Of the shown intervals, only the interval in choice B contains –2. Choices A, C, and D are incorrect and may result from either calculation errors or misidentification of the graph's x-intercepts.

Q4

14

The correct answer is 14. Let x represent the first integer and y represent the second integer. If the first integer is 5 greater than twice the second integer, then $x = 2y + 5$. It's given that the product of the two integers is 462; therefore $xy = 462$. Substituting $2y + 5$ for x in this equation yields $(2y + 5)(y) = 462$, which can be written as $2y^2 + 5y = 462$. Subtracting 462 from each side of this equation yields $2y^2 + 5y - 462 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-462)$, or -924, and whose sum is 5. The two values whose product is -924 and whose sum is 5 are 33 and -28. Thus, the equation $2y^2 + 5y - 462 = 0$ can be rewritten as $2y^2 - 28y + 33y - 462 = 0$, which is equivalent to $2y(y - 14) + 33(y - 14) = 0$, or $(2y + 33)(y - 14) = 0$. By the zero product property, it follows that $2y + 33 = 0$ or $y - 14 = 0$. Subtracting 33 from both sides of the equation $2y + 33 = 0$ yields $2y = -33$. Dividing both sides of this equation by 2 yields $y = -\frac{33}{2}$. Since y is a positive integer, the value of y isn't $-\frac{33}{2}$. Adding 14 to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting 14 for y in the equation $xy = 462$ yields $x(14) = 462$. Dividing both sides of this equation by 14 yields $x = 33$. Therefore, the two integers are 14 and 33, so the smaller of the two integers is 14.

Q5

25

The correct answer is 25. The value of $g7 - w$ is the value of gx when $x = 7 - w$, where w is a constant. Substituting $7 - w$ for x in the given equation yields $g7 - w = 7 - w7 - w - 27 - w + 6^2$, which is equivalent to $g7 - w = 7 - w5 - w13 - w^2$. It's given that the value of $g7 - w$ is 0. Substituting 0 for $g7 - w$ in the equation $g7 - w = 7 - w5 - w13 - w^2$ yields $0 = 7 - w5 - w13 - w^2$. Since the product of the three factors on the right-hand side of this equation is equal to 0, at least one of these three factors must be equal to 0. Therefore, the possible values of w can be found by setting each factor equal to 0. Setting the first factor equal to 0 yields $7 - w = 0$. Adding w to both sides of this equation yields $7 = w$. Therefore, 7 is one possible value of w. Setting the second factor equal to 0 yields $5 - w = 0$. Adding w to both sides of this equation yields $5 = w$. Therefore, 5 is a second possible value of w. Setting the third factor equal to 0 yields $13 - w^2 = 0$. Taking the square root of both sides of this equation yields $13 - w = 0$. Adding w to both sides of this equation yields $13 = w$. Therefore, 13 is a third possible value of w. Adding the three possible values of w yields $7 + 5 + 13$, or 25. Therefore, the sum of all possible values of w is 25.

Q6

-4.9

Accept also: `-49/10`

The correct answer is -4.9. The value of $f\text{-}4$ is the value of fx when $x = -4$. Substituting -4 for x in the equation $fx = 3 + \text{-}2x - x^2$ yields $f\text{-}4 = 3 + \text{-}2\text{-}4 - \text{-}4^2$, or $f\text{-}4 = 3 + \text{-}8$, which is equivalent to $f\text{-}4 = 3 + 8$, or $f\text{-}4 = 11$. Since it's given that $f\text{-}4 = c$, it follows that $c = 11$ and the value of gc is the value of $g11$. Substituting 11 for w in the equation $gw = \frac{-w}{w - 1} - w + 5$ yields $g11 = \frac{-11}{11 - 1} - 11 + 5$, or $g11 = -1.1 - 6$, which is equivalent to $g11 = 1.1 - 6$, or $g11 = -4.9$. Note that -4.9 and -49/10 are examples of ways to enter a correct answer.

Q7

A

A. The metal ball's minimum height was 3 inches above the ground.

Choice A is correct. The graph of a quadratic equation in the form $y = a(x - h)^2 + k$, where a, h, and k are positive constants, is a parabola that opens upward with vertex (h, k). The given function $f(x) = \frac{1}{9}(x - 7)^2 + 3$ is in the form $y = a(x - h)^2 + k$, where $y = f(x)$, $a = \frac{1}{9}$, $h = 7$, and $k = 3$. Therefore, the graph of $y = f(x)$ is a parabola that opens upward with vertex $(7, 3)$. Since the parabola opens upward, the vertex is the lowest point on the graph. It follows that the y-coordinate of the vertex of the graph of $y = f(x)$ is the minimum value of $f(x)$. Therefore, the minimum value of $f(x)$ is 3. It's given that $f(x) = \frac{1}{9}(x - 7)^2 + 3$ represents the metal ball's height above the ground, in inches, x seconds after it started moving on a track. Therefore, the best interpretation of the vertex of the graph of $y = f(x)$ is that the metal ball's minimum height was 3 inches above the ground.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8

C

C. $gx = \frac{6}{x+8}$

Choice C is correct. It's given that $fx = \frac{a}{x+b}$ and that the graph shown is a partial graph of $y = fx$. Substituting y for fx in the equation $fx = \frac{a}{x+b}$ yields $y = \frac{a}{x+b}$. The graph passes through the point $-7, -2$. Substituting -7 for x and -2 for y in the equation $y = \frac{a}{x+b}$ yields $-2 = \frac{a}{-7+b}$. Multiplying each side of this equation by $-7 + b$ yields $-2-7 + b = a$, or $14 - 2b = a$. The graph also passes through the point $-5, -6$. Substituting -5 for x and -6 for y in the equation $y = \frac{a}{x+b}$ yields $-6 = \frac{a}{-5+b}$. Multiplying each side of this equation by $-5 + b$ yields $-6-5 + b = a$, or $30 - 6b = a$. Substituting $14 - 2b$ for a in this equation yields $30 - 6b = 14 - 2b$. Adding $6b$ to each side of this equation yields $30 = 14 + 4b$. Subtracting 14 from each side of this equation yields $16 = 4b$. Dividing each side of this equation by 4 yields $4 = b$. Substituting 4 for b in the equation $14 - 2b = a$ yields $14 - 24 = a$, or $6 = a$. Substituting 6 for a and 4 for b in the equation $fx = \frac{a}{x+b}$ yields $fx = \frac{6}{x+4}$. It's given that $gx = fx + 4$. Substituting $x + 4$ for x in the equation $fx = \frac{6}{x+4}$ yields $fx + 4 = \frac{6}{x+4+4}$, which is equivalent to $fx + 4 = \frac{6}{x+8}$. It follows that $gx = \frac{6}{x+8}$.

Choice A is incorrect. This could define function g if $gx = fx - 4$.

Choice B is incorrect. This could define function g if $gx = fx$.

Choice D is incorrect. This could define function g if $gx = fx \cdot x + 4$.

Q9

25/4

Accept also: 6.25

The correct answer is $\frac{25}{4}$. The given equation can be rewritten in the form $fx = ax - h^2 + k$, where a, h, and k are constants. When $a > 0$, h is the value of x for which fx reaches its minimum. The given equation can be rewritten as $fx = 4x^2 - \frac{50}{4}x + 126$, which is equivalent to $fx = 4x^2 - \frac{50}{4}x + \frac{50}{8}^2 - \frac{50}{8}^2 + 126$. This equation can be rewritten as $fx = 4x - \frac{50}{8}^2 - \frac{50}{8}^2 + 126$, or $fx = 4x - \frac{50}{8}^2 - 4\frac{50}{8}^2 + 126$, which is equivalent to $fx = 4x - \frac{25}{4}^2 - \frac{121}{4}$. Therefore, $h = \frac{25}{4}$, so the value of x for which fx reaches its minimum is $\frac{25}{4}$. Note that 25/4 and 6.25 are examples of ways to enter a correct answer.

Q10 D

D. $60,000 = 50,000(1.03)^t$

Choice D is correct. Stating that the population will increase each year by 3% from the previous year is equivalent to saying that the population each year will be 103% of the population the year before. Since the initial population is 50,000, the population after t years is given by $50,000(1.03)^t$. It follows that the equation $60,000 = 50,000(1.03)^t$ can be used to estimate the number of years it will take for the population to reach 60,000.

Choice A is incorrect. This equation models how long it will take the population to decrease from 60,000 to 50,000, which is impossible given the growth factor. Choice B is incorrect and may result from misinterpreting a 3% growth as growth by a factor of 3. Additionally, this equation attempts to model how long it will take the population to decrease from 60,000 to 50,000. Choice C is incorrect and may result from misunderstanding how to model percent growth by multiplying the initial amount by a factor greater than 1.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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